Energy-Aware Task Allocation for Cooperative Lunar Exploration

Prototype Design and Evaluation Protocol

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***Abstract*: Two rovers explore a lunar test field, share visual marker detections, allocate inspections, recover unfinished work and return to base. Nearest-rover and energy-aware policies share battery feasibility constraints. Typed ROS 2 modules, camera perception and a generated Gazebo world support complete physical simulation missions. This draft reports functional validation and an evaluation protocol. Matched policy comparisons remain pending.**

***Index Terms*: multi-robot systems, lunar exploration, task allocation, ROS 2, Gazebo, energy**

I. Introduction

Multi-robot task allocation separates missions, tasks and robot capabilities [1]. Here, two rovers search separate regions and exchange detections. An inspection can be assigned to a teammate when travel cost and energy availability favour that choice.

The research question is whether energy-aware allocation improves completion and safe return compared with distance-only allocation. The contribution is an integration and evaluation protocol, without claiming optimal allocation.

II. Related Software and Scope

Space ROS supplies a VIPER-inspired rover, Mons Mouton terrain, cameras and energy plugins [2], [3], pinned at fixed revisions. The baseline uses lightweight models. An optional adapter retains upstream rover geometry and steerable joints with namespaced controllers. Its compact mission is validated; detailed terrain and physical solar integration remain pending.

ROS 2 Jazzy and Gazebo Harmonic communicate through `ros_gz`; OpenCV performs colour segmentation. Ground-truth odometry isolates allocation from localisation error. SLAM, rough terrain, mineral identification and physical solar management are outside the validated scope.

III. Mission and Architecture

The mission comprises deployment, sector exploration, visual discovery, allocated inspection and return to base. Uniquely coloured spheres represent five points of scientific interest. The rovers have different routes and initial battery fractions. Success requires every target to be confirmed and both rovers parked at their respective bases.

A. ROS Components

Each rover has a custom agent and camera detector. A central coordinator receives shared detections and status, publishes leased inspection goals and records telemetry. The agent produces velocity commands, tracks its battery, confirms nearby targets from local camera observations and handles returning and charging. A monitoring node publishes RViz markers and paths. A local mission console shows camera feeds, routes and events, with pause, resume and return controls.

Typed messages separate observations, resource state, ownership and completion. Sensor and command topics are namespaced; mission topics are shared. All mission coordinates use the world frame.

B. Visual Perception

HSV masks and contour filters detect five marker colours. Small, clipped or strongly non-circular blobs are rejected. Pinhole intrinsics map the two horizontal silhouette edges to tangent rays. For marker diameter D, their half angular separation α and mean bearing β, the estimate is

$$z_c \approx \frac{D\cos\beta}{2\sin\alpha}, \qquad y_c \approx -\frac{D\sin\beta}{2\sin\alpha}. \tag{1}$$

The camera offset and closest timestamped rover pose transform the estimate into world coordinates. This approximation assumes a mostly level field. Detection error must be measured separately against simulator truth.

An inspection requires the assigned rover to reach its stand-off, face the marker and receive recent local observations over a dwell period. A teammate's report alone cannot confirm a task.

C. Control

Agents follow sector routes and sweep cameras at survey stops. The 80 by 60 m scene includes mapped rocks, structures and crater exclusions. Bounded A* inflates hazards, marker estimates and peers by the footprint and a safety margin. A forward laser detects unlabelled obstacles. Commands are checked for curved motion and braking; missing range data stops motion. Opposing rovers pass on the right. Fixed inspection approaches avoid chasing noisy marker estimates. These safeguards assume the supplied level scene.

IV. Allocation and Energy

Each target has at most one owner, and each rover receives at most one inspection at a time. Unique target IDs suppress duplicates. Leases distinguish current assignments from expired work, and every reassignment receives a new task ID.

Let d_{ij} be rover i's approximate distance to target j's standoff, h_{ij} its estimated return distance, b_i its battery fraction and r a reserve. The synthetic energy estimate is

$$e_{ij} = k_d(d_{ij} + h_{ij}) + k_t\left(\frac{d_{ij} + h_{ij}}{v_{\text{ref}}} + t_s\right). \tag{2}$$

Both policies require $b_i \geq e_{ij} + r$. The baseline minimises d_{ij}. The energy-aware policy minimises

$$J_{ij} = d_{ij} + \lambda \frac{e_{ij} + r}{\max(b_i, \epsilon)}. \tag{3}$$

The recorded weight λ has distance units. This greedy heuristic does not solve global scheduling and may increase distance.

Energy decreases with measured distance and simulation time. Recharging at a base is accelerated for practical demonstrations. These fractions are not calibrated to a real rover. A* detours may exceed the straight-line estimate, so the agent also checks its own return budget and may release work.

A. Recovery

A missing heartbeat, unavailable or returning rover, or task timeout releases ownership. Failed tasks enter a cooldown and have bounded attempts. Results for an expired task ID are ignored. A service disables and restores a rover for repeatable failure interventions.

V. Evaluation Protocol

Run matched experiments with the same world, initial state, camera settings, mission duration, resource parameters and failure schedule. Only the policy changes. Scenarios should include nominal exploration, asymmetric batteries, temporary rover failure, and the combined energy/failure case.

Record target confirmations, success including return, simulation completion time, measured distance, synthetic energy consumption, failure-to-reassignment latency, and target-location error against simulator truth.

Retain failed runs and report success rates alongside completion times. Pair predefined layouts and random seeds. Save revisions, configuration, telemetry, events and platform details with each run.

A. Validation Boundary

Portable tests cover ownership, feasibility, recovery, detours and synthetic images. Offline rehearsals use shared mission logic, kinematic motion and oracle detections. Separate Linux gates require real rendered cameras, laser scans, odometry, assignments and full missions. Their conservative geometric footprint audit does not measure contact forces or validate physical lunar operation.

TABLE I
Functional Gazebo checks (energy policy). Gap: minimum conservative obstacle clearance; zero overlap samples.

Model and field	Time (s)	Gap (m)
Lightweight, compact	116.0	0.5421
Lightweight, expanded failure	656.5	0.2827
VIPER, compact	171.5	0.5218

VI. Functional Validation and Planned Comparison

Actual Jazzy/Harmonic runs with software rendering confirm all five targets and return both rovers (Table I). The expanded failure run disables rover 1 at world time 20 s and restores it at 50 s; its red-marker task transfers to rover 2 and is confirmed by that rover. Both lightweight runs use source revision `1317fb3`; the detailed rover uses `8bcfb9a`.

Both models pass full-circle turning, translation, rendered cameras and laser checks. Configurations, events and telemetry are retained in the project repository. These functional checks are not an allocation-policy comparison.

The policy comparison remains pending. Report paired counts, success rates, the metrics above, platform versions and a recovery timeline. Equal performance or an energy/time trade-off is a valid finding.

VII. Discussion and Limitations

Ground-truth localisation, known-size coloured markers, planar control and an accelerated battery limit external validity. Report sensitivity to the heuristic weight and error in the return-budget estimate. The central coordinator is a single point of failure. Further work includes the detailed rover on the expanded map, uneven terrain, solar charging, terrain-aware navigation and richer perception.

VIII. Conclusion

Gazebo validates compact missions with both rover models and an expanded lightweight failure mission. The design supports cooperative discovery, inspection and recovery. Allocation-policy conclusions require matched comparisons.

References

[1] B. P. Gerkey and M. J. Matarić, *A formal analysis and taxonomy of task allocation in multi-robot systems*, International Journal of Robotics Research, vol. 23, no. 9, pp. 939-954, 2004, doi: 10.1177/0278364904045564.

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